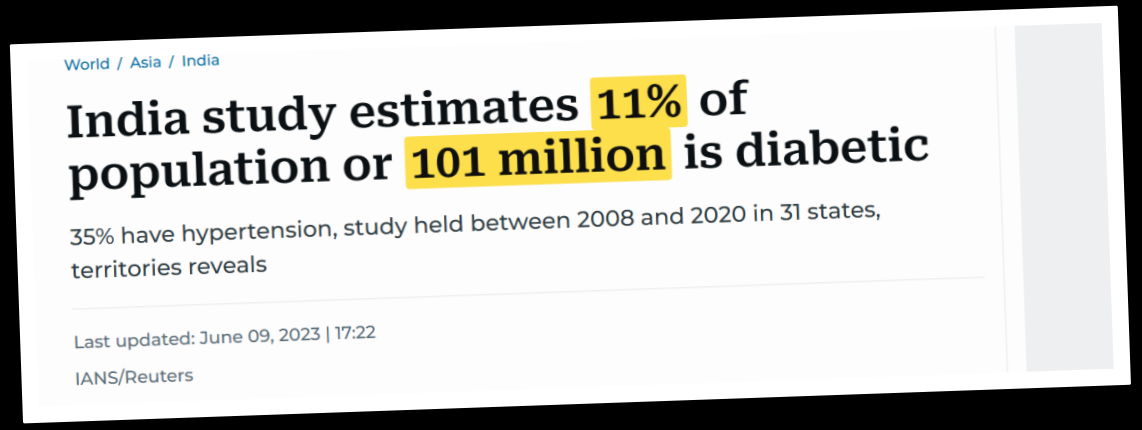
HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

See the glucose spike before it happens

PrandialTwin: a digital twin for type 2 diabetes and prediabetes in India

Team Adorable
Tharun J (team leader) · Vellore Institute of Technology, Vellore
Gargi Sharma · ABES Institute of Technology, Ghaziabad

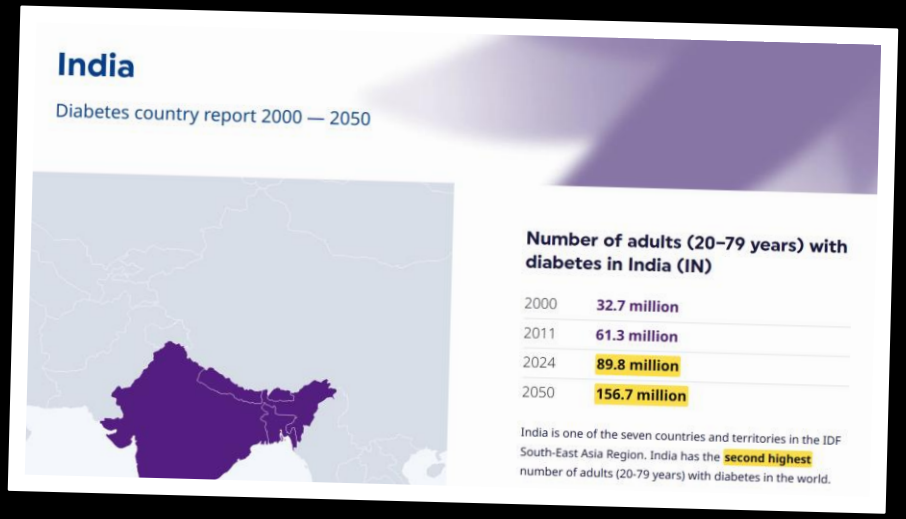
The headlines are already here



Gulf News (IANS/Reuters), 9 June 2023 [1, 2]



Imperial College London, 13 Nov 2024: NCD-RisC in The Lancet [3]



IDF Diabetes Atlas 2025, India country report [5]

> In 2021, diabetes was the direct cause of **1.6 million** deaths and **47%** of all deaths due to diabetes occurred before the age of 70 years. Another **530 000** kidney disease deaths were caused by diabetes, and high blood glucose causes around 11%

WHO diabetes fact sheet, 14 Nov 2024 [4]

Yellow marks are ours. Every number in this talk is referenced at the end.



1 in 4

Indian adults has diabetes or prediabetes

101 million with diabetes **136 million** with prediabetes

ICMR-INDIAB-17 [2]. Photo: T. Nagar, Chennai, McKay Savage from London, UK (CC BY 2.0), Wikimedia Commons

A plate built on rice, three times a day

61.6%

of the energy in the average Indian diet is **carbohydrate**

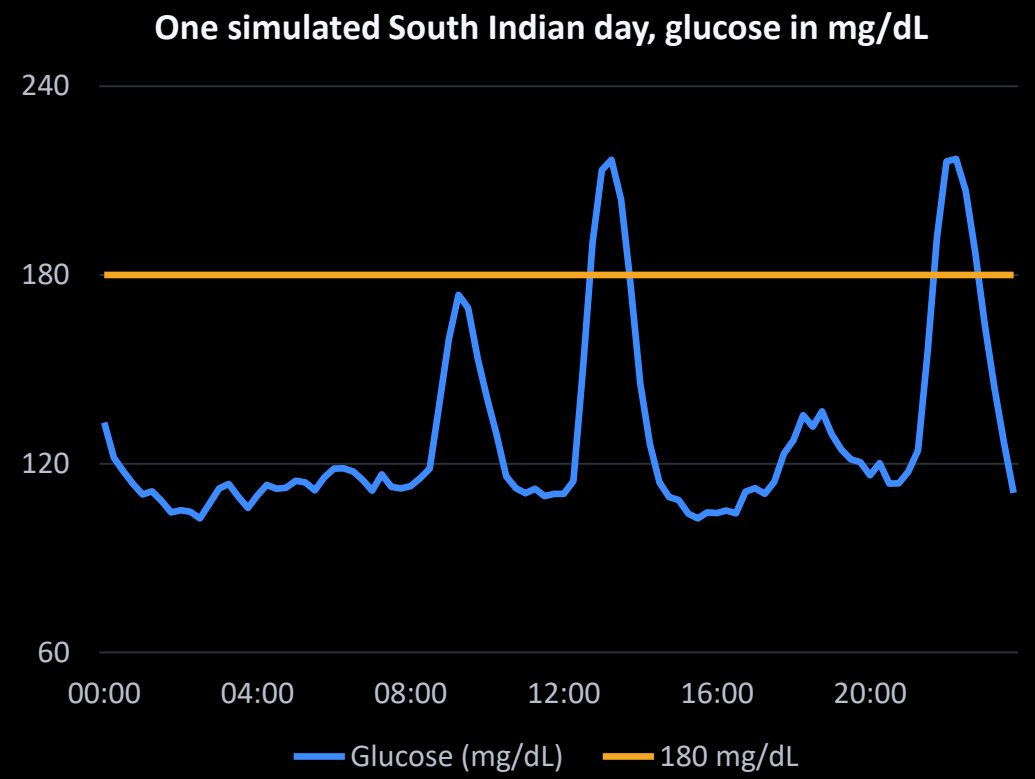
53.5%

of the time **above range** in 649 Indian patients with type 2 diabetes, on their first glucose sensor

Diet [6]; sensor study [7]. Photo: vegetarian Kerala thali, Rekhashastry (CC BY-SA 4.0), Wikimedia Commons

Diabetes care sees the average, not the spike





Meals: 08:30 rava upma · 12:15 vangi bhat, curd · 17:15 chickpea sundal · 21:10 biryani, raita

Every 3 months
an HbA1c test averages the spikes away

After 14 days
a sensor report shows them, too late to act

Missing
a warning before the meal, while a smaller portion or a walk can still help

Chart: PrandialTwin simulator, a simulated person (INS-001), not a real patient. 180 mg/dL is the international "above range" line [8]. Photo: blood glucose test, Reversing Your Diabetes Today (CC0), Wikimedia Commons

What a digital twin is, in plain words

A virtual copy of one patient's body that stays in step with them, reading by reading.

1. **It updates**
   Learns from every new glucose reading, meal and walk

2. **It predicts**
   Forecasts the next 2 to 3 hours, with an honest range

3. **It helps decide**
   Answers "what if they eat less, or walk?" and shows its own track record

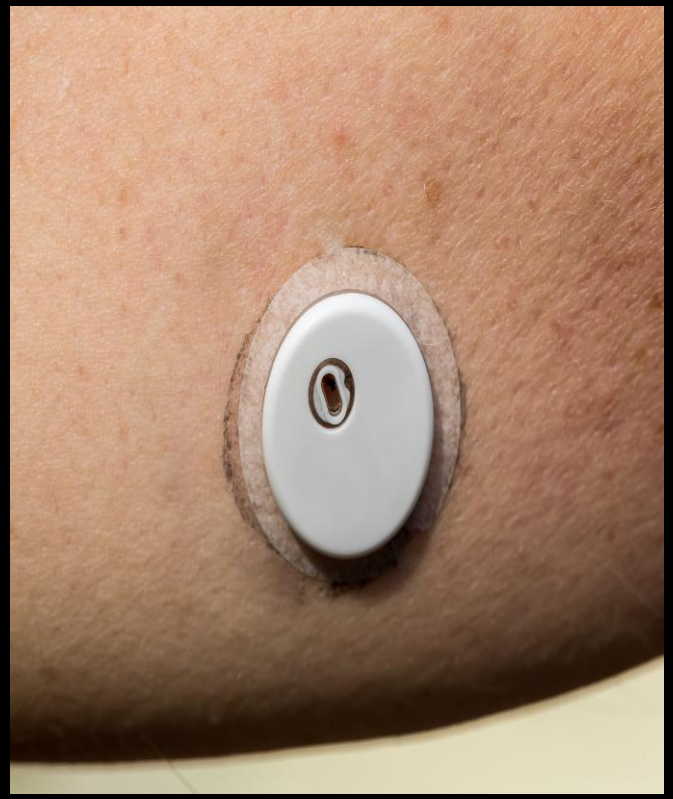
12%

of 149 published health "digital twins" do all three

PrandialTwin does all three, and tests each claim on real people

Definition [9]; review of 149 studies [10].

Three datasets, and why we trust them



A 14-day glucose sensor: the live stream

CGMacros 45 real adults, USA

Labs + two glucose sensors worn at once (Dexcom, Libre) + Fitbit + meals with macros

Why trust it: Peer-reviewed (Scientific Data 2025), PhysioNet DOI, ethics board approval, written consent [11]

ShanghaiT2DM 109 real recordings, China

15-minute sensor, clinical records with insulin doses, meals as text; never used for fitting

Why trust it: Peer-reviewed (Scientific Data 2023), figshare, MD5-checked download [12]

Synthetic patients 60 + six Indian regions × 60

Health record + glucose, heart rate, steps, sleep from one hidden physiology

Why trust it: Ground truth known, so what-ifs can be checked exactly

Fusion: record and stream belong to the same person and study window. We checksum every file (SHA-256), keep only native sensor readings, and split by patient.

Photo: FreeStyle Libre 3 sensor, Raimond Spekking (CC BY-SA 4.0), Wikimedia Commons

How our synthetic patients are made

1 Hidden severity
s ~ Beta(1.3, 1.6), from healthy to advanced type 2 diabetes

2 Physiology
insulin sensitivity, insulin release (about 15% lower per TCF7L2 risk allele), basal glucose

3 Health record
HbA1c from mean glucose (ADAG), fasting glucose and insulin, diagnoses, medications, family history

4 Daily life
meals from regional Indian plates (INDB recipes, measured GIs), walks, sleep, unlogged meals

5 Richer simulator
nonlinear stomach, first-phase insulin, liver suppression, circadian and dawn effects, sensor lag and noise

One synthetic health record (Synthea-style CSV tables) and its wearable stream

Id	Born	Sex	TCF7L2 risk alleles	Diagnosis (SNOMED)	HbA1c	Fasting glucose	Medication
SYN-001	1971	F	0	Type 2 diabetes (44054006)	6.5%	105 mg/dL	metformin

+ every 5 minutes: glucose, heart rate, steps, METs, sleep stage

6 × 60 simulated adults in six Indian regions

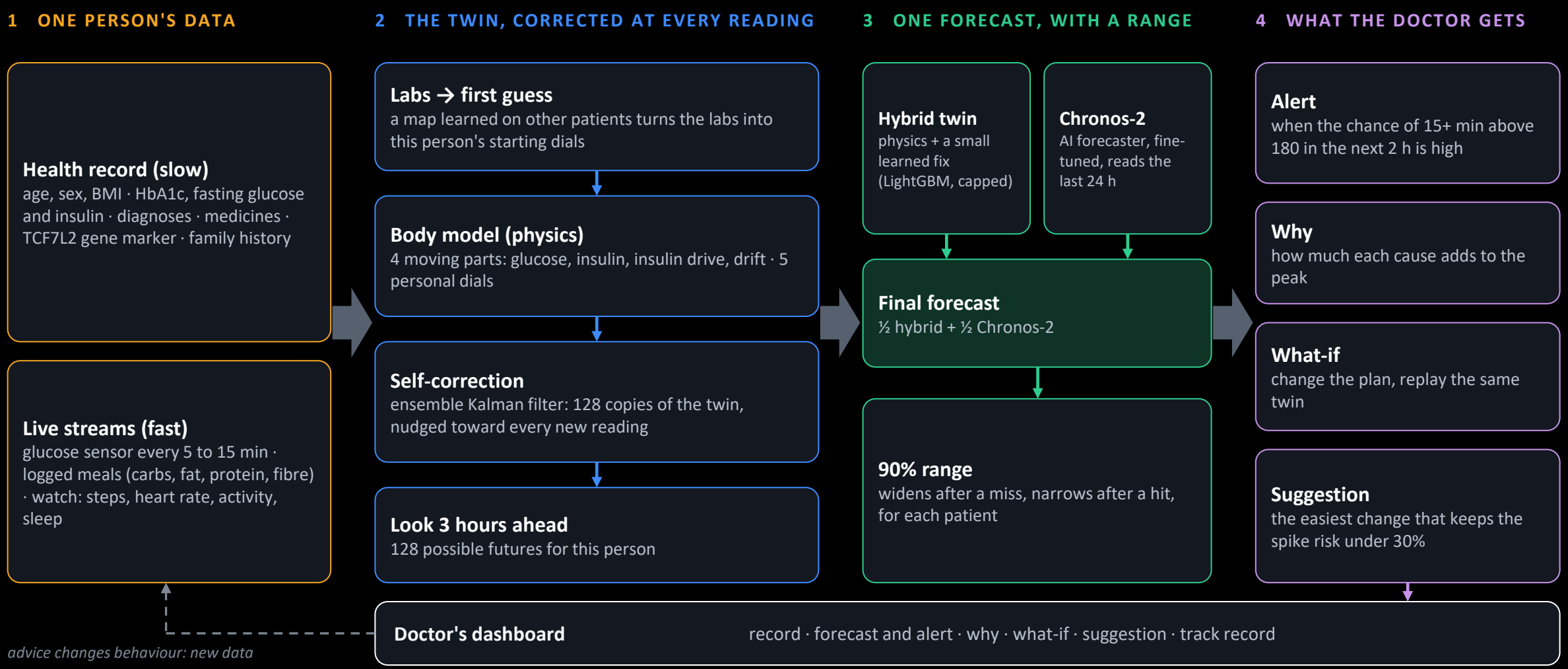
± 4.5 points every regional diet vs ICMR-INDIAB-21

± 1 point prevalence of prediabetes and diabetes

Exact true counterfactual for every what-if

Survey [13]; recipes [14]; GIs [15]; HbA1c [16]; record format [27]. Code: twin/synth.py, twin/india.py.

The architecture, in four steps



Learned beforehand, on training patients only: the labs-to-dials map, meal effects, LightGBM, the Chronos-2 fine-tune (Kaggle GPUs) and the range calibration. Checked on a simulator richer than the twin.

The physics: how blood sugar moves

$$dZ/dt = n \cdot (\varphi \cdot \max(G - G_b, 0) - Z)$$

Sugar above your normal level makes the body release insulin, Z. ϕ: how strongly

$$dX/dt = p_2 \cdot (Z - X)$$

Insulin acts with about a half-hour delay, which is why spikes happen

$$dG/dt = -(S_G + X + \beta \cdot A) \cdot G + S_G G_b + R_a(t) + D$$

Sugar in (liver, food) minus sugar out (body, insulin, activity). D: what we can't explain

$$R_a(t) = (c / BW) \cdot C \cdot k^2 \, t \, e^{-kt}$$

Food reaches the blood as a wave, up then down. Fat, protein and fibre make it slower and flatter

5 dials per person: $\theta = (G_b, \varphi, k, c, \beta)$ normal sugar level · insulin strength · digestion speed · sugar per gram of carbs · benefit of activity

Variant of the minimal model [17]. ϕ combines insulin release and sensitivity: a glucose sensor only sees their product. n, p_2, S_G are shared constants. Code: twin/core.py.

How the twin learns each person: 128 guesses

$$x = [\, G, X, Z, D, \log \theta \,]$$

4 moving parts + 5 dials; 128 copies, each with different dials

$$x_i^- = \text{model}(\, x_i \,)$$

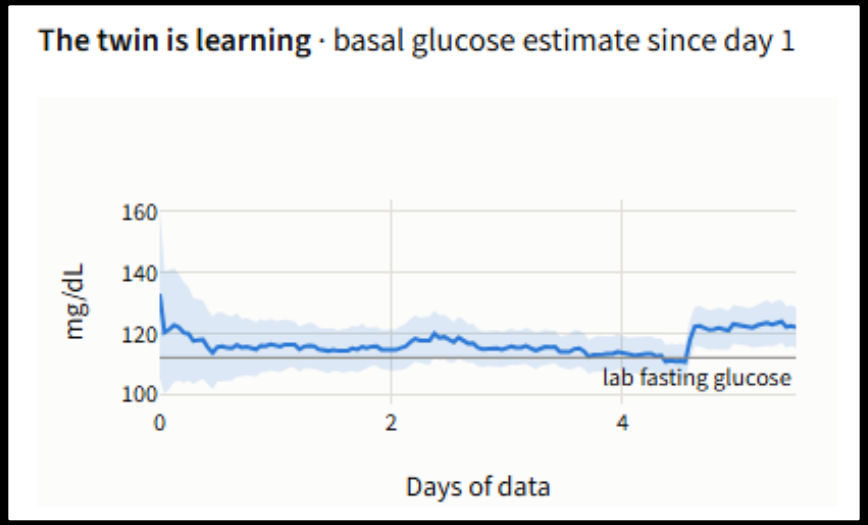
each copy predicts the next reading

$$x_i^+ = x_i^- + K \cdot (\, y + \varepsilon_i - G_i^- \,)$$

reading y arrives: pull each copy toward it, more if it was further off

$$K = \text{Cov}(\, x, G^- \,) \,/\, (\, \text{Var}(\, G^- \,) + R \,)$$

pull hard if the copies disagree, gently if the sensor is noisy (R)



From the dashboard: this patient's normal sugar level settles within a day and tracks the lab fasting glucose

A running estimate, corrected at every new reading. Never fitted to its own patient in advance

Ensemble Kalman filter [18]: moving parts and dials corrected together; it coasts through sensor gaps.

From the twin to a forecast, a range and an alert

$$\hat{y}_{\text{hybrid}} = \hat{y}_{\text{physics}} + \text{clip}(f_{\text{LGBM}}(x), \pm c)$$

physics forecast + a small learned fix, never more than c = 20 to 40 mg/dL. The physics drives; the learning only nudges

$$\hat{y} = ½\, \hat{y}_{\text{hybrid}} + ½\, \hat{y}_{\text{Chronos-2}}$$

two forecasters with different blind spots, simply averaged (fixed before testing)

$$\text{band} = [\, \hat{q}_5 - Q,\ \hat{q}_{95} + Q \,], \quad \alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{miss}_t)$$

a range that should hold 9 readings in 10. A miss widens it a little, a hit narrows it, for each patient

$$p = P(G > 180 \text{ for} \geq 15 \text{ min within 2 h}), \quad \text{alert if } p \geq \tau$$

the chance of a spike in the next 2 hours; the bell rings above τ, set for about one false alarm a day

LightGBM [19]; Chronos-2 [20]; conformalized quantile regression [21]; adaptive conformal inference [22]. Code: twin/hybrid.py.

Why, what-if, and a suggestion

Why

$$\Delta_k = \text{peak} - \text{peak without cause k}$$

remove one cause (food, activity, drift), replay, see how much the peak drops

What-if

$$\text{inputs } u \to u', \text{ same twin, same moment}$$

change the plan (a smaller portion, a walk) and replay the twin from right now

Suggestion

$$a^* = \text{argmin effort}(a) \;\; \text{s.t.} \;\; P(\text{spike} \mid a) < 0.30$$

8 easy changes (portion × walk); pick the easiest that keeps the risk under 30%



A 15-minute walk after lunch is one of the eight options the twin tries

Portion 100, 75 or 50% × walk 0, 15 or 30 min. Lifestyle options only, never medication. Photo: a park walk, Ravi Kumar (CC BY 3.0), Wikimedia Commons

Twelve forecasters, the same test

Forecaster	Kind	American (home)	Shanghai (abroad)	Indian (simulated)
Repeat last reading	baseline	30.1	29.4	31.5
LightGBM	gradient-boosted trees	25.0	28.4	15.7
GRU	recurrent network	25.1	43.2	-
LSTM	recurrent network	25.3	40.0	-
Patch transformer	transformer (PatchTST-style)	24.7	34.1	-
TS2Vec	contrastive encoder + ridge	25.9	26.8	21.0
MOMENT	foundation model, head trained	26.8	29.2	18.6
Chronos-2, zero-shot	foundation model	26.9	26.6	-
Chronos-2, fine-tuned	foundation model, fine-tuned	24.4	24.2	17.6
Physics twin	our physiology + filter	26.6	25.5	20.4
Hybrid twin	physics + capped LightGBM	24.9	26.4	**15.3**
Hybrid twin + Chronos-2 (ours)	**equal average**	**23.6**	**24.0**	15.5

Error 60 minutes ahead (RMSE, mg/dL; lower is better), same patients and moments for every model. [19, 26, 25, 24, 23, 20] 95% intervals: results/RESULTS.md, results/ROUND2.md.

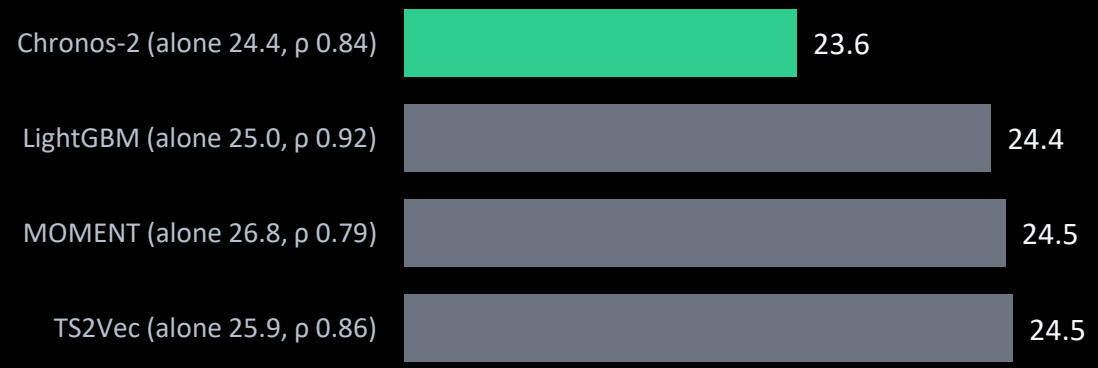
Why the twin plus Chronos-2 wins

$$\mathrm{RMSE}^2_{\mathrm{avg}} = \tfrac{1}{4}\left(\sigma_1^2 + \sigma_2^2 + 2\,\rho\,\sigma_1\,\sigma_2\right)$$

$$\sqrt{\left(\tfrac{1}{4}\left(24.9^2 + 24.4^2 + 2 \cdot 0.84 \cdot 24.9 \cdot 24.4\right)\right)} = 23.6$$

In words: an average is good when both are good (small σ) and wrong in different ways (small ρ)

Twin averaged with each partner: 60-min error at home



Accurate, and wrong differently

both are accurate, and their errors only partly line up (ρ = 0.84), so averaging cancels part of each

Abroad, the errors cancel

the twin guesses 5.4 too high, Chronos-2 3.4 too low: averaged, 24.0

Both adapt to a new person

the twin re-tunes its 5 dials at every reading; Chronos-2 re-reads the last 24 hours

σ: each forecast's own error; ρ: how alike the two forecasts' errors are (correlation, 0 to 1), measured on the 41,669 held-out forecasts.

Why the others lost

Model	How it works, in plain words	What went wrong	Evidence, 60 min unless noted
LightGBM	learns one fixed rule from one group of people	a new group: half on insulin, a slower sensor	25.0 at home, 28.4 abroad; guesses 6.5 too high
GRU, LSTM, transformer	neural networks: one rule for everyone, from a few dozen people	can't adjust to a new person; never saw insulin	24.7 to 25.3 at home, 34 to 43 abroad; 13 to 19 too high
MOMENT	reads each signal on its own, in a fixed 512-step window	never sees the meals; most of the window is empty	30 min: 20.6, the same as repeating the last reading
TS2Vec	learns to tell series apart	built to recognise, not to forecast	30 min: 18.5, against 17.2 for the hybrid
Chronos-2, zero-shot	patterns learned from many other kinds of data	has never seen how meals move glucose	26.9, then 24.4 once fine-tuned
Physics twin alone	the right shape, simplified	misses what 5 dials can't describe	26.6, then 24.9 with the small learned fix

"Too high": the average error is positive, forecasts above what happened. [23, 24, 20]

Earlier warnings, honest uncertainty

61% vs 52%

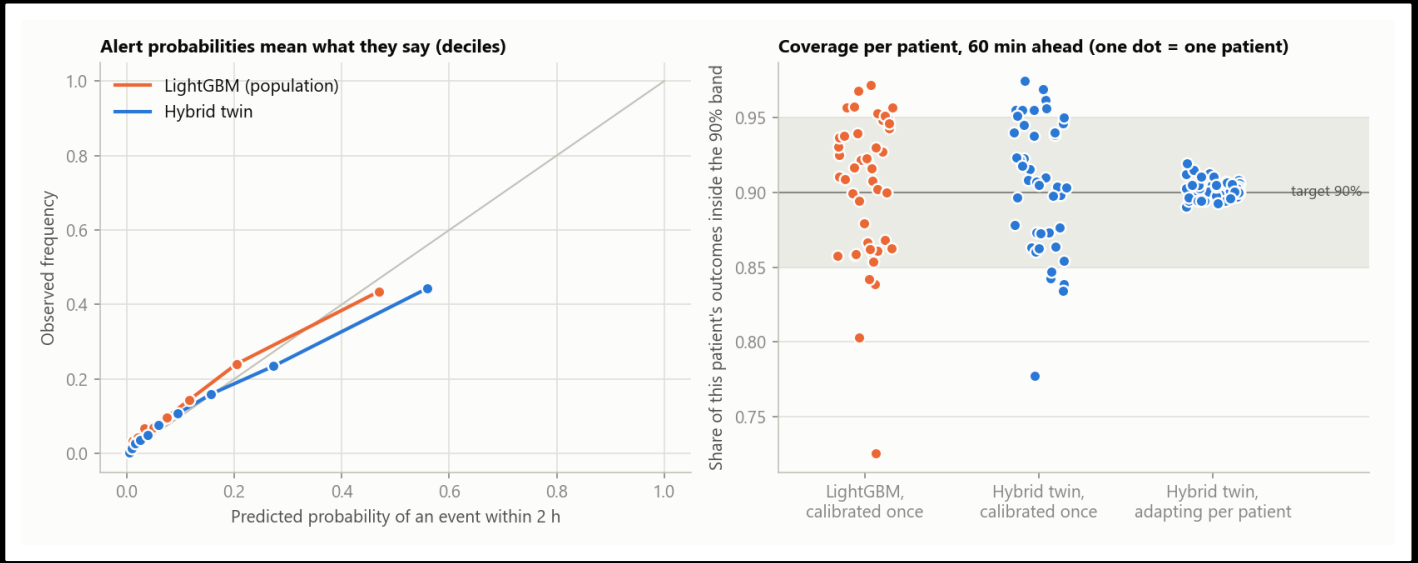
of 693 real spikes warned of before they began, twin vs standard model, at about one false alarm per patient-day

30 min

median warning before glucose stays above 180 mg/dL

0.45 vs 0.36

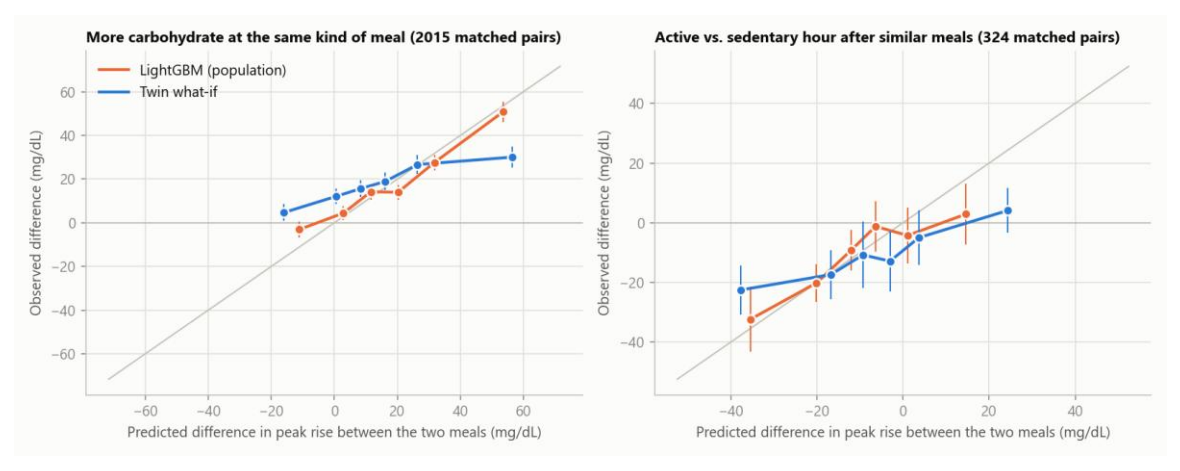
alert score on the unseen Shanghai cohort (precision-recall area; higher is better)

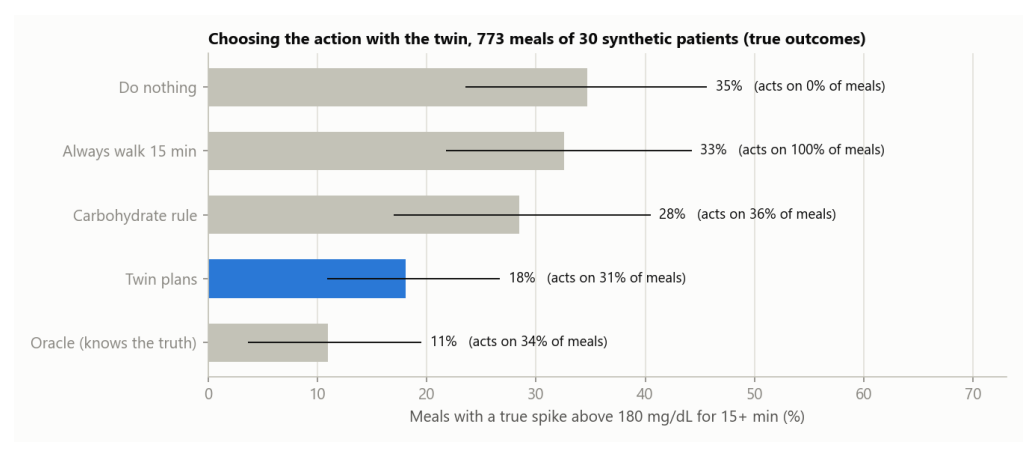


89% vs 78% of readings inside the 90% range for the worst patient: adapting per patient vs set once for everyone. Abroad: 88% vs 64%.

Alert threshold fixed on calibration patients, then run unchanged on test patients. Each dot in the right panel is one patient.

What-ifs and advice, checked





66% · 61%

direction right on real meal pairs: carbohydrate (2,015), activity (324)

−10.7 vs −9.1

active hour after a meal, mg/dL: observed vs the twin

r = 0.88

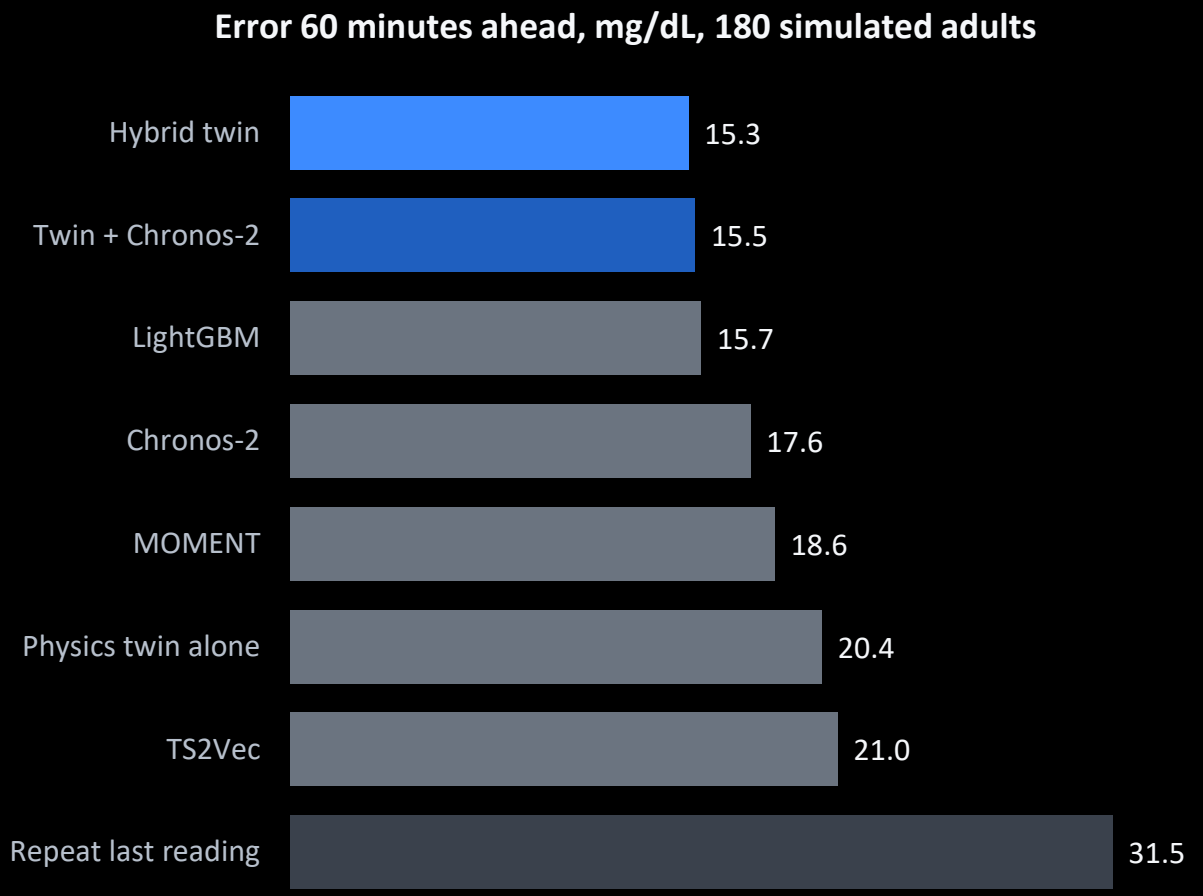
twin vs true effect of 40% less carbohydrate (synthetic truth)

35% → 18%

meals with a spike, no change vs the twin's suggestion (simulation; a carb rule: 28%)

Real pairs are matched, not randomised: a consistency check. Simulated results are an upper bound, not a trial.

Built for India: six regional plates



North
roti, dal, aloo gobhi
meals with a spike: 24%

South
rice, sambar, rasam
meals with a spike: 32%

East
rice, dalma
meals with a spike: 42%

West
rotli, dal, bhat, shaak
meals with a spike: 32%

Central
dal, chawal, roti
meals with a spike: 29%

Northeast
rice, dal, fish tenga
meals with a spike: 37%

Best in 4 of 6 regions. The simulator shares the twin's physiology: a check, not a win.

Regional diets [13], recipes [14], glycaemic indices [15]. Simulated people, not evidence about real Indians.

What did not hold up, and our limits

We hoped for	We found
Beat LightGBM at home	A tie: 24.9 vs 25.0 mg/dL at 60 min
Labs give a head start	First 24 h: −0.46 (−1.01 to 0.10). Not significant
The twin keeps improving	Updating vs frozen from day 4: −0.87 (−2.14 to 0.26). Not significant
Twin what-ifs beat LightGBM	No: 66% vs 68% direction right

Limits

- Not yet tested on real Indian patients
- Retrospective: recordings replayed, no trial
- No drug or insulin model
- Meals must be logged
- Lifestyle advice only, doctor in the loop

95% intervals by resampling patients. Full list: README, section 6.

The tech stack, all open source

The twin

Python 3.11 · NumPy · SciPy
Body model and 128-copy Kalman filter written from scratch. SciPy fits the shared dials.

Machine learning

LightGBM · scikit-learn · pandas
Capped correction, 90% range and spike alert. scikit-learn calibrates the alert probability.

AI forecasters

PyTorch · Chronos-2 · MOMENT · TS2Vec
Fine-tuned on Kaggle GPUs (2 × T4). GRU, LSTM and patch transformer baselines in PyTorch.

Doctor's dashboard

Streamlit · Plotly
Runs on a laptop. What-if sliders re-run the twin on the spot.

Data

CGMacros · ShanghaiT2DM · synthetic
Checksum-verified PhysioNet and figshare downloads, plus Synthea-style synthetic records.

Tested and reproducible

pytest · matplotlib · pinned versions
15 automated tests. One command rebuilds every number and figure in this deck.

Exact versions: requirements-lock.txt. Licences: code MIT; CGMacros CC BY-NC-SA 4.0; ShanghaiT2DM CC BY 4.0. Icons: Font Awesome (CC BY 4.0).

Open, tested, reproducible

```
> .venv\Scripts\python -I -m pytest -q
15 passed   physics · filter learning · range coverage · no future leaks · dashboard
> python -I -m twin.evaluate        every number in this talk
```

Code, notebooks, tests and this deck: github.com/iamtharunj/Adorable_Vellore_Institute_of_Technology

See the spike before it happens

Next: a 3-month hospital pilot with 100 to 200 Indian patients. We'd love a hospital partner for it.

Thank you

Team Adorable · Tharun J, VIT Vellore · Gargi Sharma, ABES Institute of Technology, Ghaziabad

github.com/iamtharunj/Adorable_Vellore_Institute_of_Technology

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